

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The best type of laser with which to do spectroscopy over a
2. True or False: The correct answer to 'The rest mass of a particle with total energy 5.0 GeV and m
3. True or False: The correct answer to 'A rod measures 1.00 m in its rest system. How fast must ar
4. True or False: The correct answer to 'A single-electron atom has the electron in the $l = 2$ state. Th
is' is '5'.
5. True or False: The correjk sue or False: The correct answer to 'Which of taf b answer to 'Which o
v following is true about any system that underhz h any system that undergoes a reversible therm
es a reversible thermodynamic process?' is 'Thmo enamic process?' is 'There are no changes in
me are no changes in the internal energy of theo internal energy of the system.'.ar"0020system.'
6. True or False: The correct answer to 'The sign of the charge carriers in a doped semiconductor c
7. True or False: The correct answer to 'The energy required to remove both electrons from the heli
8. True or False: The correct answer to 'Electromagnetic radiation emitted from a nucleus is most lik
9. True or False: The correct answer to 'A refracting telescope consists of two converging lenses se
10. True or False: The correct answer to 'Which of the following is true about any system
that undergoes a reversible thermodynamic process?' is 'There are no changes in the
internal energy of the system.'.

End of Paper